TAMPERE UNIVERSITY OF TECHNOLOGY

DEGREE CERTIFICATE

Muhammad Usman Sheikh

(Born 4th June 1983)

has passed all the examinations towards the degree of

MASTER OF SCIENCE (TECHNOLOGY)

"with distinction"

according to the International Degree Programme in

Radio Frequency Electronics

The degree consists of the following studies:

Common programme studies	20 credits	excellent
Major subject		
Radio Frequency Electronics	36 credits	very good
Minor subject		
HUOM NIMEÄ EI SAATAVILLA	30 credits	excellent
Optional Studies	7 credits	very good
Master's Thesis	30 credits	very good

"Analysis of UMTS900 deployment over GSM900/1800 and UMTS2100 network"

Examiners: Lempiäinen, Jukka
Lähdekorpi, Panu

The total scope of degree is 123 credits

Tampere, 3rd June 2009

Vice-Dean — Lauri Kettunen

Head of Degree Programme — Olli-Pekka Lundén

Certificate No. **DIA 98/2009**
Enclosed: Certificate Appendix

TAMPERE UNIVERSITY OF TECHNOLOGY

CERTIFICATE APPENDIX

Name: **Muhammad Usman S h e i k h**
Born: **4th June 1983**

Degree: **Master of Science (Technology)**
Degree programme: **International Degree Programme in Radio Frequency Electronics**
Certificate number: **DIA 98/2009**

COURSES COMPLETED	CR	CU	GRADE	DATE	LECTURER
Common programme studies	**20**				
Introductory Signal Processing	4		5	07.01.2008	Egiazarian, Karen
Algebra for linear electric systems	2		5	09.11.2007	Lehtonen, Jorma
Multivariable electromagnetic models	3		5	09.11.2007	Lehtonen, Jorma
Time-harmonic electromagnetic systems	2		5	13.12.2007	Suuriniemi, Saku
Circuits and electromagnetic fields	3		4	21.12.2007	Suuriniemi, Saku
Basic Analog Circuits	6		5	12.02.2008	Uzunov, Ivan
Radio Frequency Electronics	**36**				
Basics of RF and microwave electronics	8		5	21.05.2008	Acinas, Garzon Victor
Active RF Circuits	5		3	24.02.2009	Lunden, Olli-Pekka
RF Project	10		4	06.03.2009	Lunden, Olli-Pekka
Transmission lines and waveguides	7		5	08.05.2008	Kettunen, Lauri
Radio Propagation in Wireless Networks	3		5	26.01.2009	Turkka, Jussi
RF Equipment for Wireless Networks	3		5	03.02.2009	Lähdekorpi, Panu
HUOM NIMEÄ EI SAATAVILLA	**30**				
Design Project in Telecommunications	8		pass	16.09.2008	Koskinen, Jukka
Communication Systems Laboratory Course	5		pass	20.05.2009	Asp, Ari
Basic Course on Wireless Communications	5		5	29.01.2008	Renfors, Markku
Radio Network Planning	8		5	15.05.2008	Lempiäinen, Jukka
Basic Communication Circuits	4		4	20.05.2008	Uzunov, Ivan
Optional Studies	**7**				
Technical Writing in English	3		4	17.06.2008	Ala-Kutsi, Henna
Programming I	4		4	26.11.2007	Kilamo, Terhi
Communications engineering thesis seminar presentation			pass	18.05.2009	Renfors, Markku
Maturity Examination			pass	19.05.2009	Lempiäinen, Jukka
Master's Thesis	**30**				
Degree Thesis	30		4	03.06.2009	Lempiäinen, Jukka
"Analysis of UMTS900 deployment over GSM900/1800 and UMTS2100 network"					
Maturity Examination			pass	19.05.2009	Lempiäinen, Jukka

CERTIFICATE APPENDIX

Name: **Muhammad Usman S h e i k h**
Born: **4th June 1983**

Total credits: **123**
GPA: **Excellent**

Tampere, 3rd June 2009

Merja Majanen
Faculty Secretary

The extent of the studies is measured by credit units (CU) and credits (CR). During one academic year an average of 1600 hours work load is required equalling to 60 credits. One credit unit refers to 40 hours of academic work. Scale of grades: excellent (5), very good (4), good (3), very satisfactory (2), satisfactory (1) or pass (approved). Studies at other institutions of higher education are accepted to the degree with the grade pass (approved).

Serial No: 003645

Registration No: CIIT/FA02-BET-111/ISB

By the Authority of the Board of Governors

COMSATS Institute of Information Technology

MUHAMMAD USMAN SHEIKH s/o MUHAMMAD AKHTAR SHEIKH

Of

Islamabad Campus

has been conferred upon the degree of

Bachelor of Science in Electrical (Telecommunication) Engineering

Given on this Fifteenth day of September two thousand and Six at Islamabad

Date of Issuance: 24 April 2007

Controller of Examinations

Rector

Registrar

COMSATS Institute of Information Technology
Islamabad

1348

Transcript

Dated: 9/3/2007

Certified that **Muhammad Usman Sheikh** s/o **Muhammad Akhtar Sheikh**
Registration No. **CIIT/FA02-BET-111/ISB** Department of **Electrical Engineering**
has completed and passed all requisite courses / examinations for the degree of **Bachelor of Science in Electrical (Telecommunication) Engineering** on **31st July, 2006.**

The details of courses passed are as follows:

Course No	Semester/Course Title	Cr.	LG	GP
	Fall 2002			
ETE 1001	Introduction to Electrical Engineering	3	B	9.0
ETE 1002	Introduction to Electrical Engineering Lab	1	A	4.0
ETE 1801	Programming for Engineers	4	B	12.0
MTH 1001	Calculus I	3	A	12.0
MTH 2021	Linear Algebra	3	B-	8.1
PHY 1001	Applied Physics	4	B+	13.2
	Spring 2003			
ETE 1201	Digital Logic & Design	3	A-	11.1
ETE 1202	Digital Logic & Design Lab	1	B	3.0
ETE 1803	Programming-II	4	B-	10.8
HUM 1003	Pakistan Studies	2	B-	5.4
MTH 1002	Calculus-II	3	A	12.0
	Summer 2003			
ETE 2101	Electrical Circuits I	3	A-	11.1
ETE 2102	Electrical Circuits I Lab	1	A	4.0
	Fall 2003			
EE 353	Digital Signal Processing	3	B-	8.1
EE 353L	Digital Signal Processing Lab	1	A-	3.7
ETE 2103	Electrical Circuits II	3	C+	6.9
ETE 2104	Electrical Circuits II Lab	1	A	4.0
ETE 3103	Electronics I	3	B+	9.9
ETE 3104	Electronics I Lab	1	A	4.0
HMT 1001	Communication Skills	3	B+	9.9
MTH 2191	Differential Equations	3	A-	11.1
	Spring 2004			
ELE 2004	Probabilistic Analysis	3	A-	11.1
ELE 2208	Advance Digital Logic and Design	3	A-	11.1
ELE 2601	Optical Communications	3	B-	8.1
ELE 3104	Analog Electronics	3	C	6.0
ELE 3105	Analog Electronics Lab	1	B	3.0
ELE 3401	Systems and Controls	3	C+	6.9
ELE 3803	Engineering Economics	3	C+	6.9
	Fall 2004			
ELE 2003	Introduction to Engineering Transforms	3	C+	6.9
ELE 2503	Random Signals	3	B+	9.9
ELE 3701	Electromagnetics	3	C	6.0
ELE3205	Computer Architecture and Operating Systems	3	A-	11.1
MGT 3117	Critical Thinking and Decision Making	3	A-	11.1
	Spring 2005			
ELE 3110	Circuits and Instrumentation Lab	1	B+	3.3
ELE 3602	Communication Systems	3	B-	8.1
ELE 3702	EMT Applications	3	B	9.0
ETE 4607	Principles of Applied Telecom Technologies	3	B+	9.9
HUM 1536	Islamic Studies	2	A-	7.4
HUM 3535	Sociology and Human Behavior	3	A-	11.1
	Summer 2005			
ELE4502	DSP Applications	3	A-	11.1
	Fall 2005			
EEE- 451	Digital Communication	4	B+	13.2
EEE-443	Electrical Machines	4	B+	13.2
EEE-446	Data Communication & Comp Networks	4	A-	14.8
EEE-454	Antenna & Propagation	3	C+	6.9
MGT419	Entrepreneurship Managing	2	A	8.0
	Spring 2006			
EEE 455	Wireless Communication	4	B-	10.8
EEE 460	Digital Image Processing	4	A	16.0
ETE 4902	Final Year Design Project (A Tool for RF Planning In GSM Networks)	6	A	24.0
MGT 323	Corporate Law	3	B+	9.9

Total Credit Hours	139
* Discounted Credit Hours	-
Net Credit Hours	139 (Total GP = 448.1)
CGPA	3.22

Dy. Controller of Examinations

Errors/Omissions Excepted

(Please Turn Over)

بِسْمِ اللّٰهِ الرَّحْمٰنِ الرَّحِيْمِ

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION ISLAMABAD

Serial No. 058714 Certificate No. 633070/012714

Roll No. 553386 Registration No. 9972011170

Group PRE-ENGINEERING Attempt (s) SINGLE

HIGHER SECONDARY SCHOOL CERTIFICATE EXAMINATION
ANNUAL 2001

Certified that MUHAMMAD USMAN SHEIKH

son / daughter of MUHAMMAD AKHTAR SHEIKH

has qualified for the award of Higher Secondary School Certificate

at the examination held in the month of MAY/JUNE as a REGULAR

candidate from F G SIR SYED COLLEGE. THE MALL RAWALPINDI CANTT.

as per statement of marks given below and has obtained

grade A His / Her mark of identification : MOLE ON LIPS

SUBJECT - WISE STATEMENT OF MARKS

Sr. No	SUBJECTS	MARKS	
		Maximum	Obtained
1.	ENGLISH (COMPULSORY)	200	115
2.	URDU (COMPULSORY)	200	125
3.	PAKISTAN STUDIES	050	035
4.	ISLAMIC EDUCATION	050	046
5.	MATHEMATICS	200	151
6.	PHYSICS	200	156
7.	CHEMISTRY	200	160
8.			
	TOTAL	1100	0788

(Marks in words) SEVEN HUNDRED EIGHTY EIGHT ONLY

Islamabad Dated APRIL 24, 2003

SECRETARY

Note: This certificate is issued without alteration or erasure. (Grades over leaf)